BIOINFORMATICS

1. Introduction to Bioinformatics Databases

Bioinformatics mainly deals with **storing, organizing, retrieving, and analyzing biological data** such as:

- DNA sequences
- RNA sequences
- Protein sequences
- Protein structures
- Metabolic pathways
- Gene expression data

Because biological data is enormous, scientists store it in **specialized biological databases**.

These databases allow researchers to:

- Store biological sequences
- Retrieve information quickly
- Compare sequences
- Study protein functions
- Understand evolutionary relationships

2. Major International Sequence Databases

1. GenBank

Country: **USA**

Maintained by: **NCBI (National Center for Biotechnology Information)**

GenBank is one of the largest public databases containing:

- DNA sequences
- RNA sequences
- annotations of genes

- protein coding information

Scientists from all over the world submit newly discovered sequences to GenBank.

2. EMBL

Country: **Europe**

Maintained by: **European Bioinformatics Institute (EBI)**

EMBL database also stores nucleotide sequences submitted by researchers.

3. DDBJ

Country: **Japan**

Maintained by: **DNA Data Bank of Japan**

Important Concept: INSDC

These three databases work together through an international collaboration called:

INSDC – International Nucleotide Sequence Database Collaboration

Meaning:

GenBank (USA) EMBL (Europe) DDBJ (Japan)

All three share their data daily.

So if a sequence is submitted to one database, it becomes available in all three.

3. Primary vs Secondary Databases

Difference between Primary and Secondary databases

Primary Databases

Primary databases store **raw experimental data** submitted by scientists.

Characteristics:

- Raw sequence data
- No extensive analysis
- Direct submissions from laboratories

Examples:

- GenBank

- EMBL
- DDBJ
- PDB (protein structures)

Example:

A researcher sequences a gene in the lab and uploads the sequence to GenBank.

That sequence becomes part of the **primary database**.

Secondary Databases

Secondary databases are created by **analyzing primary database data**.

They include:

- functional annotation
- sequence patterns
- motifs
- protein families
- domain information

Characteristics:

- processed data
- curated information
- functional classification

Examples:

- PROSITE
- PRINTS
- Pfam
- BLOCKS

Simple Difference

Feature	Primary Database	Secondary Database
Data	Raw experimental data	Processed / analyzed data
Source	Direct submission	Derived from primary databases
Function	Data storage	Functional interpretation

4. Phylogenetic Trees

A phylogenetic tree shows:

Evolutionary relationships between genes, species, or proteins.

It helps answer questions such as:

- Which genes evolved from a common ancestor?
- How closely related two species are?
- How proteins evolved over time.

Example:

Human hemoglobin and chimpanzee hemoglobin are very similar → close evolutionary relationship.

5. Sequence Comparison Methods

1. Dot Matrix Method

Used to visually compare two sequences.

Steps:

1. Write one sequence on the x-axis.
2. Write another sequence on the y-axis.
3. Place dots where characters match.

Diagonal lines indicate similarity.

Used to detect:

- repeats
- inversions
- insertions
- deletions

2. FASTA

FASTA is a **sequence alignment algorithm**.(**FAST-All**)

Purpose:

To compare a query sequence with sequences in a database.

It identifies:

- similar DNA sequences
- similar protein sequences

FASTA performs **global alignment**.

Meaning it compares the entire sequence.

3. BLAST

BLAST = **Basic Local Alignment Search Tool**

One of the most widely used tools in bioinformatics.

Purpose:

Find similar sequences in a database.

Difference from FASTA:

FASTA	BLAST
Global alignment	Local alignment
Whole sequence	Similar regions

BLAST is much faster.

6. Difference Between Bioinformatics and Computational Biology

Bioinformatics

Focuses on:

- storing biological data
- organizing data
- analyzing biological sequences

Examples:

- database management
- genome annotation
- sequence analysis

Computational Biology

Focuses on:

- mathematical modeling
- simulation of biological processes
- understanding biological systems

Examples:

- protein folding models
- metabolic pathway simulations
- systems biology modeling

7. First Protein Studied

Hemoglobin

Hemoglobin was one of the first proteins whose sequence and structure were studied extensively.

Reasons:

- abundant protein
- easy to isolate

- medically important

Hemoglobin studies helped scientists understand:

- protein structure
- oxygen transport
- genetic mutations like sickle cell anemia

8. Margaret O. Dayhoff

Important scientist in bioinformatics.

Contributions:

- Created the first **protein sequence database**
- Developed **protein substitution matrices**
- Created the **Atlas of Protein Sequence and Structure**

She is considered a **pioneer of bioinformatics**.

9. First Nucleotide Sequence Database

The first major nucleotide sequence database:

GenBank

Hosted by **NCBI**

It stores DNA and RNA sequences submitted by researchers worldwide.

10. Protein Databases

Protein databases store information about protein sequences and functions.

These are divided into:

Primary Protein Databases

1. **Swiss-Prot**

Now part of **UniProtKB**

Features:

- manually curated
- high-quality annotations

- protein function information

2. **TrEMBL**

Contains automatically annotated protein sequences.

3. **PIR (Protein Information Resource)**

One of the earliest protein sequence databases.

11. Secondary Protein Databases

These classify proteins based on **families, motifs, and domains**.

Examples:

PROSITE

Detects:

- protein motifs
- functional sites

PRINTS

Contains **protein fingerprints**

Fingerprint = group of conserved motifs.

BLOCKS

Database of conserved regions in proteins.

Pfam

Database of **protein domains and families**.

Uses **Hidden Markov Models**.

12. Protein Structural Databases

These store **3D structures of proteins**.

PDB – Protein Data Bank

The most important structural database.

Contains:

- 3D protein structures
- nucleic acid structures
- complexes

Structures are determined using:

- X-ray crystallography
- NMR spectroscopy
- Cryo-EM

SCOP

Structural Classification of Proteins.

Classifies proteins based on structural relationships.

Hierarchy:

Class Fold Superfamily Family

CATH

Another structural classification database.

CATH stands for:

C – Class A – Architecture T – Topology H – Homologous superfamily

13. Enzyme Databases

These databases store information about enzymes.

BRENDA

Full form:

Braunschweig Enzyme Database

Contains:

- enzyme functions
- metabolic data
- enzyme structures
- enzyme kinetics

ExPASy

Expert Protein Analysis System.

Developed by:

Swiss Institute of Bioinformatics.

Provides:

- protein analysis tools
- enzyme information
- protein structure tools

14. Clinical Databases

Clinical databases store genetic disease information.

HGMD

Human Gene Mutation Database.

Contains information about:

- disease-causing mutations
- genetic disorders

OMIM

Online Mendelian Inheritance in Man.

Contains:

- human genes
- inherited diseases
- genetic disorders

15. KEGG Database

KEGG = **Kyoto Encyclopedia of Genes and Genomes**

Purpose:

To understand biological systems such as:

- metabolic pathways

- disease pathways
- cellular processes

Example:

KEGG pathway can show:

Gene → protein → pathway → disease

16. Bibliographic Databases

These databases help find scientific research papers.

Example:

PubMed

Contains millions of biomedical research articles.

Used for:

- literature search
- research references

17. Data Organization in Biological Databases

Biological databases must organize data efficiently.

Two important concepts:

Data

Raw biological information.

Example:

DNA sequence

ATGCGTAC

Metadata

Information about the data.

Example:

- organism name
- gene name

- sequencing method
- publication

18. Redundancy and Indexing

Redundancy

Occurs when the same sequence appears multiple times in a database.

This increases database size unnecessarily.

Solution:

Remove duplicate entries.

Indexing

Indexing improves **data retrieval speed**.

Example:

Like a book index.

Instead of searching every page, you directly jump to the relevant section.

19. Sequence Formats (PIR)

Important features:

- header lines start with **semicolon (;)**
- sequence data follows the header
- sequences are stored in standardized format

20. Data Retrieval Systems

Tools used to retrieve biological data.

Entrez

Developed by **NCBI**.

Web-based system that integrates multiple databases.

Allows searching across:

- GenBank

- Protein database
- PubMed
- GEO
- Structure databases

DBGET

Database retrieval system used in KEGG.

SRS – Sequence Retrieval System

Used to retrieve sequences from multiple biological databases.

21. Entrez Identifiers

Every entry in Entrez has a unique identifier called:

Accession number

Example:

NM_000518

This allows easy retrieval of sequences.

22. Related Databases Mentioned

MMDB

Molecular Modeling Database

Contains:

- 3D biomolecular structures

GEO

Gene Expression Omnibus.

Stores gene expression data.

23. BLAST as a Retrieval Tool

BLAST helps retrieve biological information by finding similar sequences.

Process:

1. Input query sequence
2. BLAST searches database
3. Finds similar sequences
4. Displays alignment results

This helps identify:

- gene function
- evolutionary relationships
- homologous proteins

24. Challenges in Retrieving Biological Data

Some difficulties include:

- massive data volume
- redundant entries
- incomplete annotations
- inconsistent data formats

Bioinformatics tools solve these problems.

25. Important Exam Questions

Likely questions include:

1. Difference between primary and secondary databases.
2. Explain BLAST and FASTA.
3. Explain Swiss-Prot and TrEMBL.
4. Describe protein structural databases.
5. What is KEGG?
6. Explain Entrez retrieval system.

All Questions 1

Difference between Primary and Secondary Databases

Primary Databases

Primary databases store **raw experimental biological data** submitted directly by researchers.

Characteristics:

- Contain **original sequence data**
- Minimal analysis
- Direct lab submissions
- Frequently updated

Examples:

- **GenBank**
- **EMBL**
- **DDBJ**
- **PDB**

Example: If a scientist sequences a gene and uploads it to GenBank, the sequence becomes part of a **primary database**.

Secondary Databases

Secondary databases are created by **analyzing and organizing information from primary databases**.

They contain:

- Functional annotations
- Protein motifs
- Protein families
- Structural classifications

Examples:

- **PROSITE**
- **PRINTS**
- **Pfam**
- **BLOCKS**

2. Difference Between Bioinformatics and Computational Biology

Bioinformatics

Bioinformatics focuses on **storage, organization, and analysis of biological data using computers**.

Activities include:

- DNA sequence analysis
- Database management
- Genome annotation
- Sequence comparison

Examples of tools:

- BLAST
- FASTA
- GenBank
- UniProt

Computational Biology

Computational biology focuses on **modeling biological systems using mathematics, algorithms, and simulations**.

Activities include:

- Protein folding simulations
- Systems biology modeling
- Drug interaction modeling
- Evolutionary simulations

3. How Does BLAST Function as a Retrieval Tool?

BLAST stands for **Basic Local Alignment Search Tool**.

It is used to **search biological databases for similar sequences**.

Working Steps

Step 1 — Input Query Sequence

A DNA or protein sequence is submitted to BLAST.

Example:

ATGCCGATGAC

Step 2 — Database Search

BLAST compares the query sequence with millions of sequences stored in databases like:

- GenBank
- Protein databases

Step 3 — Find Local Similarities

BLAST identifies **similar regions (local alignments)** between sequences.

Step 4 — Ranking Results

Results are ranked based on:

- similarity score
- E-value
- alignment length

Why BLAST is Useful

BLAST helps scientists:

- identify unknown genes
- predict protein function
- find evolutionary relationships
- detect homologous sequences

4. What Challenges Are Faced While Retrieving Biological Data?

Retrieving biological data presents several challenges.

1. Large Data Volume

Modern sequencing technologies produce **massive amounts of biological data**.

Example: Human genome contains **3 billion base pairs**.

2. Data Redundancy

The same sequence may appear multiple times in databases.

This increases:

- storage requirements
- retrieval complexity

3. Data Heterogeneity

Biological data comes from many sources:

- genomes
- proteomes
- expression data
- clinical records

Different formats make integration difficult.

4. Incomplete Annotations

Many sequences in databases are poorly annotated or lack functional information.

5. Rapid Data Growth

New biological data is generated faster than it can be curated.

5. How Are Flat File Databases Organized?

A flat file database stores data in **simple text files**.

Each entry contains **fields arranged sequentially**.

Structure of Flat File Databases

Each record contains multiple fields:

Example structure:

ID Sequence Identifier DE Description OS Organism SQ Sequence

Characteristics

- Simple format
- Human readable
- Easy to distribute
- Used in early bioinformatics databases

Examples

- GenBank flat file format
- EMBL format
- Swiss-Prot format

6. What is Metadata and Why Is It Important?

Metadata Definition

Metadata is **data about data**.

It provides descriptive information about biological sequences.

Example

DNA Sequence:

ATGCGTACGT

Metadata may include:

- organism name
- gene name
- chromosome location
- sequencing method
- author information

Importance of Metadata

Metadata helps in:

- efficient data searching
- proper data classification
- understanding biological context
- improving database organization

7. What is Indexing and How Does It Improve Retrieval Efficiency?

Indexing Definition

Indexing is a technique used to **organize database information so that data can be retrieved quickly**.

Example

In a book, an index helps locate topics quickly instead of searching every page.

Similarly, in biological databases:

- sequences
- gene names
- accession numbers

are indexed.

Benefits of Indexing

1. Faster search
2. Reduced computation time
3. Efficient data retrieval
4. Better database performance

8. What is the Role of NCBI?

NCBI stands for:

National Center for Biotechnology Information

Location: USA

Functions of NCBI

NCBI provides:

Biological Databases

- GenBank
- PubMed
- GEO

- Protein database
- Structure database

Bioinformatics Tools

NCBI developed important tools such as:

- BLAST
- Entrez
- Genome browser

Data Integration

NCBI integrates multiple biological databases into one platform.

Importance

NCBI acts as a **central hub for global biological data**.

Scientists worldwide use NCBI resources.

9. Justify the Statement:

"Entrez is a Database Retrieval System"

Entrez is a **web-based search and retrieval system developed by NCBI**.

It allows users to search across multiple databases simultaneously.

Databases Accessible Through Entrez

- GenBank
- PubMed
- Protein database
- Genome database
- GEO
- Structure database

Features of Entrez

1. Integrated searching across databases
2. Cross-linking between data types
3. Easy web interface
4. Unique identifiers for entries

Example

A gene entry in Entrez may link to:

Gene sequence ↓ Protein structure ↓ Research papers ↓ Disease information

10. What Are Entrez Identifiers?

Every entry in Entrez has a **unique identifier**.

These identifiers help retrieve information quickly.

Types of Identifiers

Accession Number

Example:

NM_000546

Used for gene sequences.

GI Number (older system)

Used previously to identify sequence entries.

Purpose

Identifiers allow:

- easy referencing
- accurate data retrieval
- database cross-linking

11. What Are Entrez Links and Neighbors?

Entrez provides **links and neighbors** to connect related biological information.

Entrez Links

Links connect entries across different databases.

Example:

Gene → Protein → Structure → Literature

Entrez Neighbors

Neighbors show **similar or related entries**.

Example:

- similar genes
- homologous proteins
- related sequences

Importance

These features help researchers explore biological relationships easily.

Summary of Questions From Your Notes

The questions found across your notebook pages were:

1. Difference between primary and secondary databases
2. Difference between bioinformatics and computational biology
3. How does BLAST function as a retrieval tool?
4. What challenges are faced in retrieving biological data?
5. How are flat file databases organized?
6. What is metadata and why is it important?
7. What is indexing and how does it improve retrieval efficiency?
8. What is the role of NCBI?
9. Justify the statement "Entrez is a database retrieval system"
10. What are Entrez identifiers?
11. What are Entrez links and neighbors?

If you want, I can also **do one extremely useful thing for you**: